

Solution	Marks	
3. (a) Using integration by parts: $\int e^{-x}[f(x) - f'(x)] dx$ $= \int e^{-x} f(x) dx - \int e^{-x} f'(x) dx$ $= \int e^{-x} f(x) dx - (e^{-x} f(x) - \int f(x) d(e^{-x}))$ $= \int e^{-x} f(x) dx - e^{-x} f(x) - \int e^{-x} f(x) dx$ $= -e^{-x} f(x) + C$ (b) R.H.S. = $\frac{1}{x+x^n} - \frac{d}{dx} \left(\frac{1}{x+x^n} \right)$ $= \frac{1}{x+x^n} - \left(-\frac{1+nx^{n-1}}{(x+x^n)^2} \right)$ $= \frac{x+x^n+1+nx^{n-1}}{(x+x^n)^2}$ $= \frac{x^n+nx^{n-1}+x+1}{(x+x^n)^2} = \text{L.H.S.}$ (c) For $n=3$, let $f(x) = \frac{1}{x+x^3}$. By (b), $\frac{x^3+3x^2+x+1}{(x+x^3)^2} = f(x) - f'(x)$. By (a), $\int e^{-x}[f(x) - f'(x)] dx = -e^{-x} f(x) + C$. $\int_5^8 e^{-x} \cdot \frac{x^3+3x^2+x+1}{(x+x^3)^2} dx = \left[-e^{-x} \cdot \frac{1}{x+x^3} \right]_5^8$ $= -\frac{e^{-8}}{8+8^3} - \left(-\frac{e^{-5}}{5+5^3} \right)$ $= \frac{e^{-5}}{130} - \frac{e^{-8}}{520} \quad \left(\text{or } \frac{4-e^{-3}}{520e^5} \right)$	1M 1A 1M 1A 1M 1A	

Solution	Marks	
4. (a) $\ \vec{v}_1\ = \sqrt{4^2 + 3^2} = 5$ $\vec{v}_1 \cdot \vec{v}_3 = \ \vec{v}_1\ \ \vec{v}_3\ \cos \theta \implies 5 \ \vec{v}_3\ \cos \theta = \cos \theta \implies \ \vec{v}_3\ = \frac{1}{5}$ Since $\vec{v}_1 \cdot \vec{v}_2 = 0$, $\vec{v}_2 \perp \vec{v}_1$. Since $\vec{v}_2 \cdot \vec{v}_3 > 0$, the angle between $\vec{v}_2$ and $\vec{v}_3$ is $\left(\frac{\pi}{2} - \theta\right)$. $\vec{v}_2 \cdot \vec{v}_3 = \ \vec{v}_2\ \ \vec{v}_3\ \cos\left(\frac{\pi}{2} - \theta\right) = \frac{1}{5} \ \vec{v}_2\ \sin \theta$ (b) When $\ \vec{v}_2\ = \frac{1}{5}$ and $\theta = \frac{\pi}{6}$: $\ \vec{v}_2\ = \ \vec{v}_3\ = \frac{1}{5}$ Angle between $\vec{v}_2$ and $\vec{v}_3 = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$ $\ \vec{v}_2 - \vec{v}_3\ ^2 = \ \vec{v}_2\ ^2 + \ \vec{v}_3\ ^2 - 2\vec{v}_2 \cdot \vec{v}_3$ $= \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 - 2\left(\frac{1}{5}\right)\left(\frac{1}{5}\right)\cos\frac{\pi}{3}$ $= \frac{1}{25}$ $\therefore$ Distance $= \ \vec{v}_2 - \vec{v}_3\ = \frac{1}{5}$	1M 1M 1A 1M 1M 1A	

Solution	Marks	
5. (a) $\left(x + \frac{1}{x}\right)^n (ax - 4)^2 = x^{-n}(x^2 + 1)^n(a^2x^2 - 8ax + 16)$ $\mu_2 = 16C_1^n + a^2C_0^n = 16n + a^2 = 84$ $\mu_3 = -8aC_1^n = -8an = -80 \implies an = 10 \implies a = \frac{10}{n}$ $16n + \left(\frac{10}{n}\right)^2 = 84 \implies 4n^3 - 21n^2 + 25 = 0$ $(n - 5)(4n^2 - n - 5) = 0 \implies (n - 5)(4n - 5)(n + 1) = 0$ $\therefore n$ is a positive integer $\therefore n = 5, a = 2$ (b) $\left(2x + \frac{1}{x}\right)^2 (x - 4)^5 = \left(4x^2 + 4 + \frac{1}{x^2}\right) \sum_{r=0}^5 C_r^5 x^{5-r} (-4)^r$ Constant term: $= 4 \cdot C_5^5 (-4)^5 + 1 \cdot C_3^5 (-4)^3$ $= 4(1)(-1024) + 1(10)(-64)$ $= -4096 - 640$ $= -4736$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
6. (a) Let $P(n)$ be the statement "$\sum_{k=1}^n (5 + 4k) \cdot 3^{2k-1} = \frac{3^{2n}(24n + 27) - 27}{16}$". For $n = 1$: L.H.S. $= (5 + 4(1)) \cdot 3^{2(1)-1} = 9 \cdot 3 = 27$ R.H.S. $= \frac{3^2(24(1) + 27) - 27}{16} = \frac{9(51) - 27}{16} = \frac{432}{16} = 27 = \text{L.H.S.} \therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m (5 + 4k) \cdot 3^{2k-1} = \frac{3^{2m}(24m + 27) - 27}{16}$. For $n = m + 1$: L.H.S. $= \sum_{k=1}^{m+1} (5 + 4k) \cdot 3^{2k-1}$ $= \frac{3^{2m}(24m + 27) - 27}{16} + [5 + 4(m + 1)] \cdot 3^{2(m+1)-1}$ $= \frac{3^{2m}(24m + 27) - 27 + 16(4m + 9) \cdot 3^{2m+1}}{16}$ $= \frac{3^{2m}(24m + 27) + 48(4m + 9) \cdot 3^{2m} - 27}{16}$ $= \frac{3^{2m}[24m + 27 + 192m + 432] - 27}{16}$ $= \frac{3^{2m}[216m + 459] - 27}{16}$ $= \frac{3^{2m} \cdot 9[24(m + 1) + 27] - 27}{16}$ $= \frac{3^{2(m+1)}[24(m + 1) + 27] - 27}{16} = \text{R.H.S.}$ $\therefore P(m + 1)$ is true if $P(m)$ is true. By the principle of mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{k=10}^{90} (5 + 4k) \cdot 3^{2k+1} = 3^2 \sum_{k=10}^{90} (5 + 4k) \cdot 3^{2k-1}$ $= 9 \left(\sum_{k=1}^{90} (5 + 4k) \cdot 3^{2k-1} - \sum_{k=1}^9 (5 + 4k) \cdot 3^{2k-1} \right)$ $= 9 \left(\frac{3^{180}(24(90) + 27) - 27}{16} - \frac{3^{18}(24(9) + 27) - 27}{16} \right)$ $= 9 \left(\frac{2187 \cdot 3^{180} - 243 \cdot 3^{18}}{16} \right)$ $= \frac{19683 \cdot 3^{180} - 2187 \cdot 3^{18}}{16} \quad \left(\text{or } \frac{3^{187} - 3^{25}}{16} \right)$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
7. (a) For C_2: $x^2 + (y - 1)^2 = \frac{3}{4} \implies x^2 = \frac{3}{4} - (y - 1)^2$ C_1 and C_2 intersect at $y = \frac{1}{2}$: $x^2 = \frac{1}{2}$ (from C_1) and $x^2 = \frac{3}{4} - \left(-\frac{1}{2}\right)^2 = \frac{1}{2}$ (from C_2). Note that for $0 \leq y \leq \frac{1}{2}$, C_2 has $y \leq 1$, so $y - 1 \leq 0 \implies y - 1 = -\sqrt{\frac{3}{4} - x^2}$. The bottom of the sphere C_2 reaches $y = 1 - \frac{\sqrt{3}}{2} \approx 0.134 > 0$. Thus, for $0 \leq y < 1 - \frac{\sqrt{3}}{2}$, the cavity is bounded only by C_1. For $0 \leq y \leq 1 - \frac{\sqrt{3}}{2}$: $x_{C_1}^2 = y$ For $1 - \frac{\sqrt{3}}{2} \leq y \leq \frac{1}{2}$: $x_{C_2}^2 = -y^2 + 2y - \frac{1}{4}$, $x_{C_1}^2 = y$ $V_{\text{cavity}} = \pi \int_0^{1/2} y \, dy - \pi \int_{1-\frac{\sqrt{3}}{2}}^{1/2} \left(\frac{3}{4} - (y-1)^2 \right) dy$ $= \pi \left[\frac{y^2}{2} \right]_0^{1/2} - \pi \left[-\frac{y^3}{3} + y^2 - \frac{y}{4} \right]_{1-\frac{\sqrt{3}}{2}}^{1/2}$ $= \pi \left(\frac{1}{8} \right) - \pi \left(\frac{1}{12} - \left(\frac{5}{12} - \frac{\sqrt{3}}{4} \right) \right)$ $= \frac{(11 - 6\sqrt{3})\pi}{24}$ (b)(i) Case 1: $0 < h \leq 1 - \frac{\sqrt{3}}{2}$ $V(h) = \pi \int_0^h y \, dy = \frac{\pi h^2}{2}$ Case 2: $1 - \frac{\sqrt{3}}{2} < h \leq \frac{1}{2}$ $V(h) = \pi \int_0^h y \, dy - \pi \int_{1-\frac{\sqrt{3}}{2}}^h \left(\frac{3}{4} - (y-1)^2 \right) dy$ $= \pi \left(\frac{h^3}{3} - \frac{h^2}{2} + \frac{h}{4} + \frac{5 - 3\sqrt{3}}{12} \right)$ (b)(ii) Since $h = \frac{1}{4}$ and $1 - \frac{\sqrt{3}}{2} \approx 0.134 < \frac{1}{4} < \frac{1}{2}$: We use $V(h) = \pi \left(-\frac{h^3}{3} + \frac{h^2}{2} - \frac{h}{4} + \frac{3\sqrt{3} - 5}{12} \right)$ for $h > 1 - \frac{\sqrt{3}}{2}$. $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt} = \pi \left(-h^2 + h - \frac{1}{4} \right) \frac{dh}{dt}$ Given $\frac{dV}{dt} = \frac{\pi}{1280}$ and $h = \frac{1}{4}$:	1M 1M 1A 1A 1A 1M	

Solution		Marks
$\frac{\pi}{1280} = \pi \left(-\left(\frac{1}{4}\right)^2 + \frac{1}{4} - \frac{1}{4} \right) \frac{dh}{dt}$ $\frac{\pi}{1280} = -\frac{\pi}{16} \cdot \frac{dh}{dt}$ $\frac{dh}{dt} = -\frac{1}{80} \text{ units/s}$		1A

Solution	Marks	
8. (a) $\begin{vmatrix} -2 & 2 & 2 - \lambda_2 \\ -\lambda_1 & 1 & 1 + \lambda_1 \\ 3 & -1 & \lambda_2 - 1 \end{vmatrix} = 0$ L.H.S. $= -2(\lambda_2 - 1 + 1 + \lambda_1) - 2(-\lambda_1\lambda_2 + \lambda_1 - 3 - 3\lambda_1) + (2 - \lambda_2)(\lambda_1 - 3)$ $= \lambda_1\lambda_2 + 4\lambda_1 + \lambda_2$ $= (\lambda_1 + 1)(\lambda_2 + 4) = 0$ $\therefore \lambda_1, \lambda_2$ are non-negative integers. $\therefore \lambda_1 = 0$ and $\lambda_2 = 0$ $\therefore \lambda_1 + \lambda_2 = 0$ (b) Substitute $\lambda_1 = 0, \lambda_2 = 0$: $\begin{cases} 3x + 2y + 2z = k \\ y + z = k^2 \\ -2x - y - z = 1 \end{cases}$ $\Delta_y = \begin{vmatrix} 3 & k & 2 \\ 0 & k^2 & 1 \\ -2 & 1 & -1 \end{vmatrix} = k^2 - 2k - 3 = 0$ $(k - 3)(k + 1) = 0 \implies k = 3 \text{ or } k = -1$ (c) Case 1: $k = 3$ $\begin{cases} 3x + 2y + 2z = 3 \\ y + z = 9 \\ -2x - y - z = 1 \end{cases} \implies \begin{cases} y + z = 9 \\ x = -5 \end{cases}$ Let $z = t$ ($t \in \mathbb{R}$). $\therefore (x, y, z) = (-5, 9 - t, t)$ Case 2: $k = -1$ $\begin{cases} 3x + 2y + 2z = -1 \\ y + z = 1 \\ -2x - y - z = 1 \end{cases} \implies \begin{cases} y + z = 1 \\ x = -1 \end{cases}$ Let $z = t$ ($t \in \mathbb{R}$).	1M 1M 1A 1M 1A 1A	

Solution	Marks	
$\therefore (x, y, z) = (-1, 1 - t, t)$	1A	

Solution	Marks																
SECTION B																	
9.																	
(a) $f(x) = \frac{x^3 + x^2 - 5x + 3}{x^2 + 3} = x + 1 + \frac{-8x}{x^2 + 3}$ Since $x^2 + 3 > 0$ for all real x , there are no vertical asymptotes. $\lim_{x \rightarrow \pm\infty} [f(x) - (x + 1)] = \lim_{x \rightarrow \pm\infty} \frac{-8x}{x^2 + 3} = 0$ $\therefore$ The oblique asymptote of C is $y = x + 1$.	1M																
(b) $f'(x) = 1 + \frac{-8(x^2 + 3) - (-8x)(2x)}{(x^2 + 3)^2}$ $= 1 + \frac{8(x^2 - 3)}{(x^2 + 3)^2}$ $= \frac{x^4 + 14x^2 - 15}{(x^2 + 3)^2}$ $= \frac{(x^2 - 1)(x^2 + 15)}{(x^2 + 3)^2}$ Set $f'(x) = 0 \implies x = 1, -1$. $f''(x) = \frac{16x(x^2 + 3)^2 - 8(x^2 - 3) \cdot 2(x^2 + 3)(2x)}{(x^2 + 3)^4}$ $= \frac{16x(x^2 + 3) - 32x(x^2 - 3)}{(x^2 + 3)^3}$ $= \frac{48x(3 - x^2)}{(x^2 + 3)^3}$ Set $f''(x) = 0 \implies x = 0, \sqrt{3}, -\sqrt{3}$.	1M																
<table><tr><td>x</td><td>$(-\infty, -\sqrt{3})$</td><td>$-\sqrt{3}$</td><td>$(-\sqrt{3}, 0)$</td><td>0</td><td>$(0, \sqrt{3})$</td><td>$\sqrt{3}$</td><td>$(\sqrt{3}, \infty)$</td></tr><tr><td>$f''(x)$</td><td>$+$</td><td>0</td><td>$-$</td><td>0</td><td>$+$</td><td>0</td><td>$-$</td></tr></table> Since $f''(x)$ changes sign across $x = -\sqrt{3}, 0$, and $\sqrt{3}$: $f(0) = 1$ $f(-\sqrt{3}) = \frac{-3\sqrt{3} + 3 + 5\sqrt{3} + 3}{6} = 1 + \frac{\sqrt{3}}{3}$ $f(\sqrt{3}) = \frac{3\sqrt{3} + 3 - 5\sqrt{3} + 3}{6} = 1 - \frac{\sqrt{3}}{3}$ $\therefore$ The points of inflexion are $(0, 1)$, $\left(-\sqrt{3}, 1 + \frac{\sqrt{3}}{3}\right)$, and $\left(\sqrt{3}, 1 - \frac{\sqrt{3}}{3}\right)$.	x	$(-\infty, -\sqrt{3})$	$-\sqrt{3}$	$(-\sqrt{3}, 0)$	0	$(0, \sqrt{3})$	$\sqrt{3}$	$(\sqrt{3}, \infty)$	$f''(x)$	$+$	0	$-$	0	$+$	0	$-$	1A
x	$(-\infty, -\sqrt{3})$	$-\sqrt{3}$	$(-\sqrt{3}, 0)$	0	$(0, \sqrt{3})$	$\sqrt{3}$	$(\sqrt{3}, \infty)$										
$f''(x)$	$+$	0	$-$	0	$+$	0	$-$										
(c) Sketching C : - Correct shape showing turning points at $x = \pm 1$ and points of inflexion at $x = 0, \pm\sqrt{3}$. - Oblique asymptote $y = x + 1$ clearly drawn. - Correct intercepts: y-intercept $(0, 1)$, x-intercepts at $(-3, 0)$ and $(1, 0)$.	3A																
(d)(i) Slope of $L = f'(-\sqrt{3}) = \frac{8(3 - 3)}{(3 + 3)^2} + 1 = 1$. Equation of tangent L :																	

Solution	Marks	
$y - \left(1 + \frac{\sqrt{3}}{3}\right) = 1 \cdot (x - (-\sqrt{3}))$	1M	
$y = x + \frac{4\sqrt{3}}{3} + 1$	1A	
(d)(ii) Consider $L(x) - f(x)$:		
$L(x) - f(x) = \left(x + 1 + \frac{4\sqrt{3}}{3}\right) - \left(x + 1 - \frac{8x}{x^2 + 3}\right)$ $= \frac{4\sqrt{3}}{3} + \frac{8x}{x^2 + 3}$ $= \frac{4\sqrt{3}(x^2 + 3) + 24x}{3(x^2 + 3)}$ $= \frac{4\sqrt{3}(x^2 + 2\sqrt{3}x + 3)}{3(x^2 + 3)}$ $= \frac{4\sqrt{3}(x + \sqrt{3})^2}{3(x^2 + 3)}$	1M	
Since $(x + \sqrt{3})^2 \geq 0$ and $x^2 + 3 > 0$ for all real numbers x ,	1M	
$L(x) - f(x) \geq 0 \implies L \geq f(x)$ for all real numbers x .	1A	

Solution	Marks	
10. (a)(i) $\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx$ $= x \ln x - x + C$ (a)(ii) $\int_{-a}^a f(x) \, dx = \int_{-a}^0 f(x) \, dx + \int_0^a f(x) \, dx$ For the first integral, let $u = -x \implies dx = -du$. When $x = -a, u = a$; when $x = 0, u = 0$. $\int_{-a}^0 f(x) \, dx = \int_a^0 f(-u)(-du) = \int_0^a f(-u) \, du$ Since $f(-u) = f(u)$, $\int_0^a f(-u) \, du = \int_0^a f(u) \, du = \int_0^a f(x) \, dx$. $\therefore \int_{-a}^a f(x) \, dx = \int_0^a f(x) \, dx + \int_0^a f(x) \, dx = 2 \int_0^a f(x) \, dx$. (b)(i) $I(t) = \int_{1/t}^t \frac{\ln\left(\frac{1+x^2}{x^2}\right)}{x(1+x^2)} \, dx$ Let $u = \frac{1}{x} \implies x = \frac{1}{u}, dx = -\frac{1}{u^2} \, du$. When $x = 1/t, u = t$ when $x = t, u = 1/t$. $I(t) = \int_t^{1/t} \frac{\ln\left(\frac{1+(1/u)^2}{(1/u)^2}\right)}{\frac{1}{u}\left(1+\frac{1}{u^2}\right)} \left(-\frac{1}{u^2}\right) \, du$ $= \int_{1/t}^t \frac{u \ln(1+u^2)}{1+u^2} \, du$ $= \int_{1/t}^t \frac{x \ln(1+x^2)}{1+x^2} \, dx$ (b)(ii) Let $x = e^y \implies dx = e^y \, dy$. When $x = 1/t, y = -\ln t$; when $x = t, y = \ln t$. $I(t) = \int_{-\ln t}^{\ln t} \frac{e^y \ln(1+e^{2y})}{1+e^{2y}} e^y \, dy$ $= \int_{-\ln t}^0 \frac{e^{2y} \ln(1+e^{2y})}{1+e^{2y}} \, dy + \int_0^{\ln t} \frac{e^{2y} \ln(1+e^{2y})}{1+e^{2y}} \, dy$ Let $y = -w \implies dy = -dw$. $\int_{-\ln t}^0 \frac{e^{2y} \ln(1+e^{2y})}{1+e^{2y}} \, dy$ $= \int_{\ln t}^0 \frac{e^{-2w} \ln(1+e^{-2w})}{1+e^{-2w}} \, dw$ $= \int_0^{\ln t} \frac{\ln(1+e^{2w}) - \ln(e^{2w})}{1+e^{2w}} \, dw$ $= \int_0^{\ln t} \frac{\ln(1+e^{2w}) - 2w}{1+e^{2w}} \, dw$ $= \int_0^{\ln t} \frac{\ln(1+e^{2y}) - 2y}{1+e^{2y}} \, dy$	1M 1A 1M 1A 1M 1A 1M 1M	

Solution	Marks	
$\begin{aligned}\therefore I(t) &= \int_0^{\ln t} \frac{\ln(1 + e^{2y}) - 2y}{1 + e^{2y}} dy + \int_0^{\ln t} \frac{e^{2y} \ln(1 + e^{2y})}{1 + e^{2y}} dy \\ &= \int_0^{\ln t} \frac{(1 + e^{2y}) \ln(1 + e^{2y}) - 2y}{1 + e^{2y}} dy \\ &= \int_0^{\ln t} \left(\ln(1 + e^y) - \frac{2y}{1 + e^{2y}} \right) dy \\ &= \int_0^{\ln t} \left(\ln(1 + e^{2x}) - \frac{2x}{1 + e^{2x}} \right) dx\end{aligned}$	1A	
(b)(iii) For $t = e$: $\begin{aligned}\int_0^1 \left(\ln(1 + e^{2x}) - \frac{2x}{1 + e^{2x}} \right) dx &= [x \ln(1 + e^{2x})]_0^1 - \int_0^1 x \cdot \frac{2e^{2x}}{1 + e^{2x}} dx - \int_0^1 \frac{2x}{1 + e^{2x}} dx \\ &= \ln(1 + e^2) - \int_0^1 2x dx \\ &= \ln(1 + e^2) - [x^2]_0^1 \\ &= \ln(1 + e^2) - 1\end{aligned}$	1M	
$\therefore \int_{1/e}^e \frac{\ln\left(\frac{1+x^2}{x^2}\right)}{x(1+x^2)} dx = \ln(1 + e^2) - 1.$	1A	

Solution	Marks	
11.		
(a)(i) $\det(\lambda I - A) = \det \begin{pmatrix} \lambda - 10 & 12 \\ -4 & \lambda + 4 \end{pmatrix} = (\lambda - 10)(\lambda + 4) + 48 = \lambda^2 - 6\lambda + 8 = 0$	1M	
$(\lambda - 4)(\lambda - 2) = 0 \implies \lambda_1 = 4, \lambda_2 = 2$	1A	
(a)(ii) $B = A - 2I = \begin{pmatrix} 8 & -12 \\ 4 & -6 \end{pmatrix}$		
$C = A - 4I = \begin{pmatrix} 6 & -12 \\ 4 & -8 \end{pmatrix}$	1M	
$BC = \begin{pmatrix} 8 & -12 \\ 4 & -6 \end{pmatrix} \begin{pmatrix} 6 & -12 \\ 4 & -8 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{0}$	1A	
(b)(i) $B^2 = \begin{pmatrix} 8 & -12 \\ 4 & -6 \end{pmatrix} \begin{pmatrix} 8 & -12 \\ 4 & -6 \end{pmatrix} = \begin{pmatrix} 16 & -24 \\ 8 & -12 \end{pmatrix} = 2 \begin{pmatrix} 8 & -12 \\ 4 & -6 \end{pmatrix} = 2B$	1M+1A	
(b)(ii) $B^2 = 2B$ $B^3 = B^2B$ $= (2B)B$ $= 2(B^2)$ $= 4B$ $\therefore B^n = B^{n-1}B$ $= (2^{n-2}B)B$ $= 2^{n-2}(B^2)$ $= 2^{n-1}B$	1M	
$\therefore \mu = 2^{n-1}.$	1A	
(c)(i) $pB + qC = p(A - 2I) + q(A - 4I) = (p + q)A - (2p + 4q)I = I$	1M	
Setting $p + q = 0 \implies q = -p. -(2p + 4(-p)) = 1 \implies 2p = 1 \implies p = \frac{1}{2}. \therefore q = -\frac{1}{2}.$	1A	
(c)(ii) $(pB + qC)A^n = IA^n = A^n. \frac{1}{2}BA^n - \frac{1}{2}CA^n = A^n.$		
Note that $A = B + 2I = C + 4I. BA = B(B + 2I) = B^2 + 2B = 4B \implies BA^n = 4^n B = 2^{2n}B.$ $CA = C(C + 4I) = C^2 + 4C = 2C \implies CA^n = 2^n C.$	1M	
$\therefore A^n = \frac{1}{2}(2^{2n}B) - \frac{1}{2}(2^n C) = 2^{2n-1}B - 2^{n-1}C.$	1A	
(c)(iii) $X_n = \sum_{k=1}^n A^k = \sum_{k=1}^n (2^{2k-1}B - 2^{k-1}C) = (\sum_{k=1}^n 2^{2k-1})B - (\sum_{k=1}^n 2^{k-1})C$ $\sum_{k=1}^n 2^{2k-1} = \frac{2(4^n - 1)}{3}, \quad \sum_{k=1}^n 2^{k-1} = 2^n - 1$		

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$X_n = \frac{2(4^n - 1)}{3} \begin{pmatrix} 8 & -12 \\ 4 & -6 \end{pmatrix} - (2^n - 1) \begin{pmatrix} 6 & -12 \\ 4 & -8 \end{pmatrix}$ $= \begin{pmatrix} \frac{2}{3}(8 \cdot 2^n - 1)(2^n - 1) & -4(2 \cdot 2^n - 1)(2^n - 1) \\ \frac{4}{3}(2 \cdot 2^n - 1)(2^n - 1) & -4(2^n - 1)^2 \end{pmatrix}$ $\therefore \det(X_n) = \left[\frac{2}{3}(8 \cdot 2^n - 1)(2^n - 1)\right] \left[-4(2^n - 1)^2\right] - \left[-4(2 \cdot 2^n - 1)(2^n - 1)\right] \left[\frac{4}{3}(2 \cdot 2^n - 1)(2^n - 1)\right]$ $= \frac{8}{3}(2^n - 1)[(2^n)^2 - 1]$ Let $2^n = u$. $120 = \frac{8}{3}(2^n - 1)[(2^n)^2 - 1]$ $120 = \frac{8}{3}(u - 1)(u^2 - 1)$ $45 = (u - 1)(u^2 - 1)$ $3^2 \cdot 5 = (u - 1)^2(u + 1)$ $\therefore 2^n = 4 \implies n = 2$	1M 1A	

Solution	Marks	
12. (a)(i) $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3) = \mathbf{v}_1 \times \mathbf{v}_1 - \mathbf{v}_1 \times \mathbf{v}_3 - \mathbf{v}_2 \times \mathbf{v}_1 + \mathbf{v}_2 \times \mathbf{v}_3$ $= \mathbf{0} + \mathbf{v}_3 \times \mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2 + \mathbf{v}_2 \times \mathbf{v}_3 = \mathbf{v}_1 \times \mathbf{v}_2 + \mathbf{v}_2 \times \mathbf{v}_3 + \mathbf{v}_3 \times \mathbf{v}_1$ (a)(ii) If V_1, V_2, V_3, V_4 are coplanar, $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3) = (\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_4)$ $\therefore (\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3) \neq (\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_4)$ Therefore, V_1, V_2, V_3, V_4 are not coplanar. (b)(i) $\mathbf{v}_1 - \mathbf{v}_2 = -\cos\theta\mathbf{i} - \sin\theta\mathbf{j} + \mathbf{k}$ $\mathbf{v}_1 - \mathbf{v}_3 = \cos\theta\mathbf{i} - \sin\theta\mathbf{j} + \mathbf{k}$ $\mathbf{v}_1 - \mathbf{v}_4 = 2\mathbf{k}$ $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\cos\theta & -\sin\theta & 1 \\ \cos\theta & -\sin\theta & 1 \end{vmatrix} = -2\sin\theta\mathbf{i} + 2\cos\theta\mathbf{j} + 2\sin\theta\cos\theta\mathbf{k}$ $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_4) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\cos\theta & -\sin\theta & 1 \\ 0 & 0 & 2 \end{vmatrix} = -2\sin\theta\mathbf{i} + 2\cos\theta\mathbf{j}$ Since the $\mathbf{k}$-component $2\sin\theta\cos\theta \neq 0$ for $0 < \theta < \frac{\pi}{2}$, they are not equal. (b)(ii) Let the angle between $\mathbf{v}_2, \mathbf{v}_3$ be ϕ. Since $(\mathbf{v}_2 - \mathbf{v}_3) \perp \mathbf{v}_1, \mathbf{v}_4$. $\therefore V = 2 \times \left[\frac{1}{3} \cdot \left(\frac{1}{2} \mathbf{v}_2 \mathbf{v}_3 \sin\phi \cdot 1 \right) \right]$ $= 2 \times \left[\frac{1}{3} \cdot \left(\frac{1}{2} \mathbf{v}_2 \times \mathbf{v}_3 \cdot 1 \right) \right]$ $\mathbf{v}_2 \times \mathbf{v}_3 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\theta & \sin\theta & 0 \\ -\cos\theta & \sin\theta & 0 \end{vmatrix} = 2\cos\theta\sin\theta\mathbf{k} = \sin 2\theta\mathbf{k}$ $\therefore V = \frac{1}{3} \sin 2\theta$ (c)(i) Since $0 < \theta < \frac{\pi}{2} \implies 0 < 2\theta < \pi$, $\sin 2\theta$ attains its maximum value 1 when $2\theta = \frac{\pi}{2} \implies \theta = \frac{\pi}{4}$. $\therefore$ Maximum $V = \frac{1}{3}$. (c)(ii) When $\theta = \frac{\pi}{4}$: Let $\mathbf{n}_1 = (\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3), \mathbf{n}_2 = (\mathbf{v}_4 - \mathbf{v}_2) \times (\mathbf{v}_4 - \mathbf{v}_3)$	1M 1A 1A 1M 1A 1M 1A	

